Three segments

When pins are not on the same row or column, an L is not your only pattern

The idea

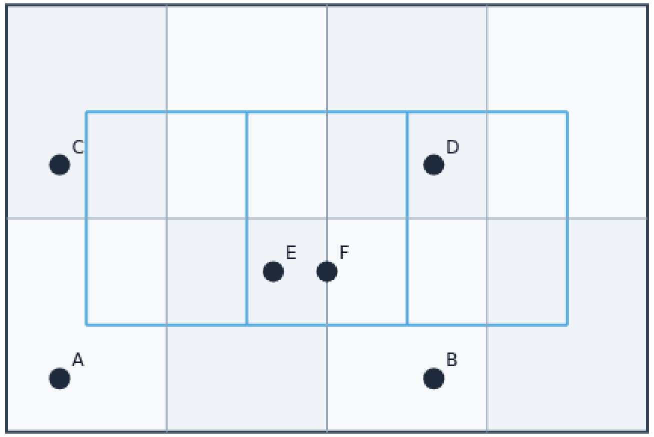
- If a and b share a row or column, fall back to L-route
- Otherwise with prefer HZ pick mid column as the average of the two columns, walk H–V–H
- Path_to_edges should show at least three edges for off-axis pairs like zero comma zero to

Two bends

Z-PATTERN ROUTE

Two bends

Step 1 / 5 · z · module02-03-pattern-z-route



Explanation

Z-route uses a midpoint column (or row) with two corners.

- vs one-bend L
- More edges possible

```
Toy Z
```

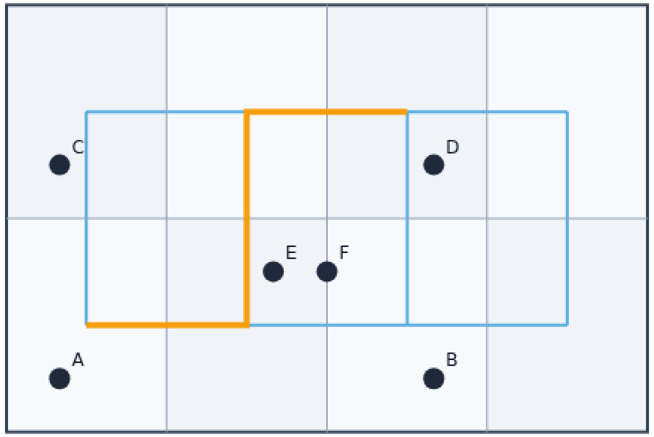
Two bends

A–D Z path

Z-PATTERN ROUTE

A–D Z path

Step 2 / 5 · ab · module02-03-pattern-z-route



Explanation

On spread, Z from A to D shows the mid-column bend.

- Compare to L
- Same pins

```
Highlight path
```

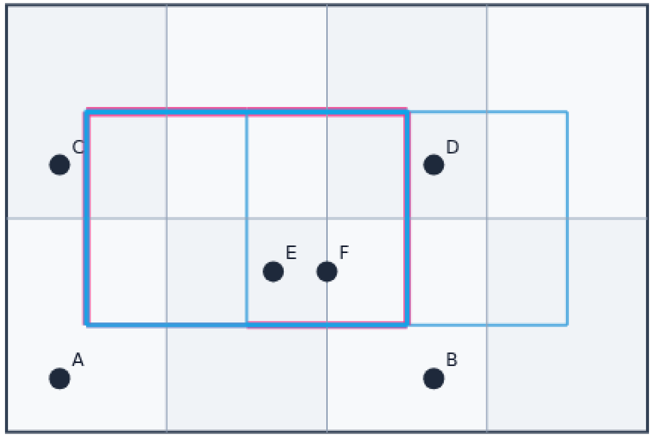
A–D Z path

L reference

Z-PATTERN ROUTE

L reference

Step 3 / 5 · l · module02-03-pattern-z-route



Explanation

L-HV is the fast pattern route baseline.

- Prefer HV/VH
- Pattern router

```
Baseline
```

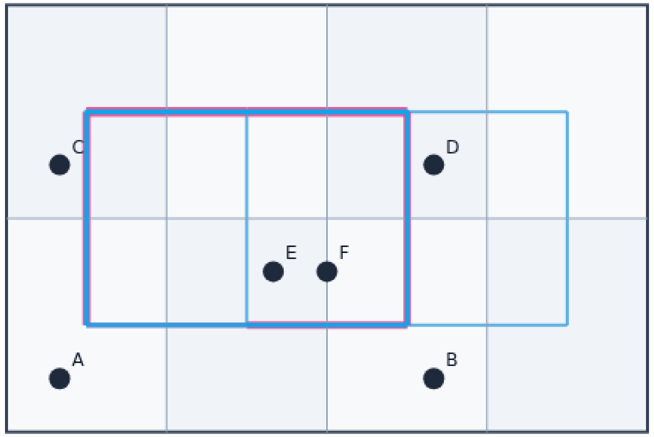
L reference

Overflow tradeoff

Z-PATTERN ROUTE

Overflow tradeoff

Step 4 / 5 · ov · module02-03-pattern-z-route



Explanation

Extra bends can spread or concentrate edge usage.

- Not always better
- Measure

```
Cap=2
```

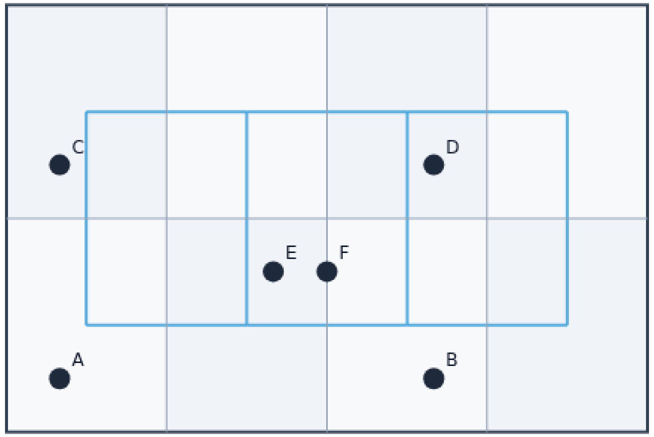
Overflow tradeoff

Maze escape

Z-PATTERN ROUTE

Maze escape

Step 5 / 5 · next · module02-03-pattern-z-route



Explanation

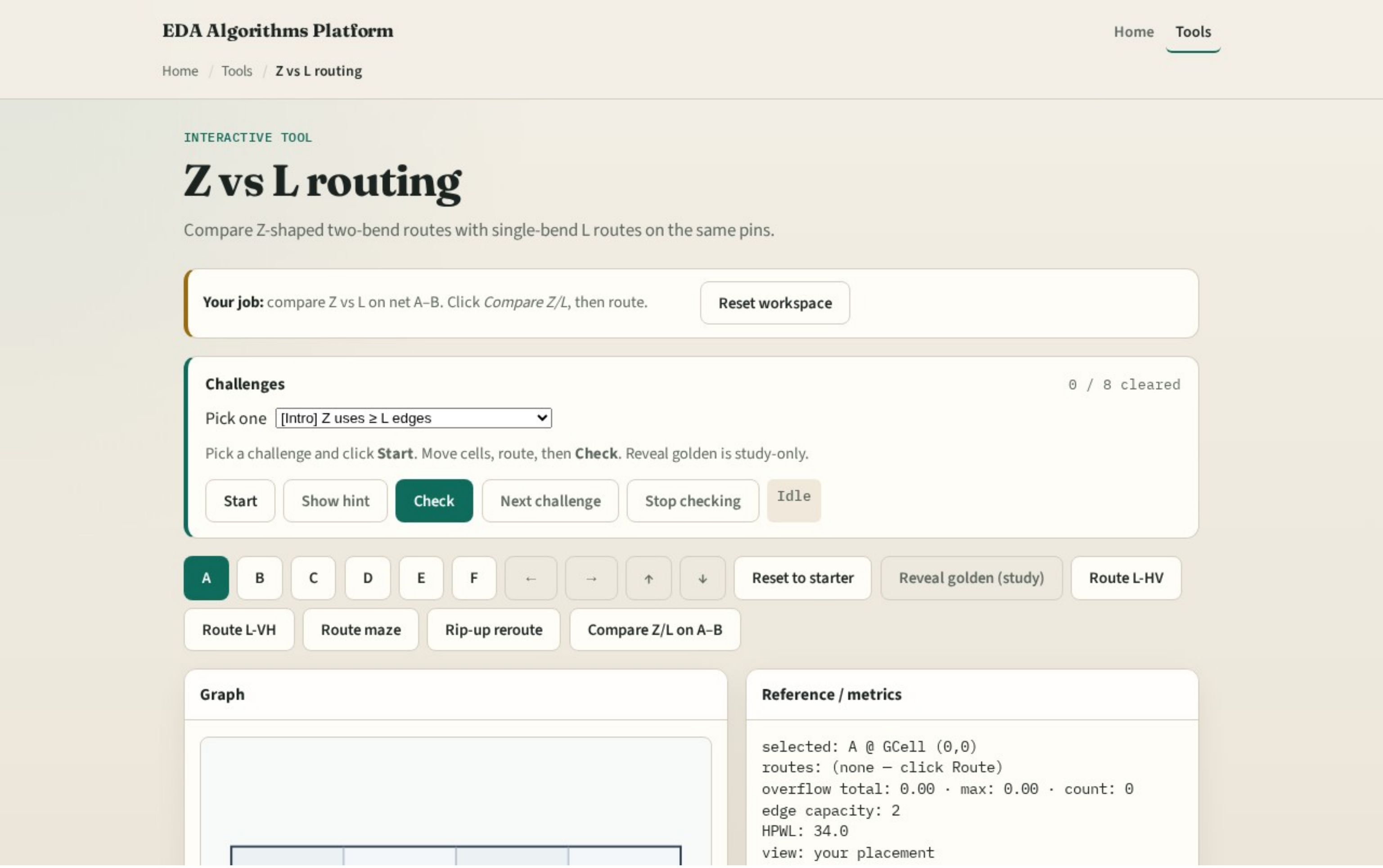
When L/Z overflow, maze search detours.

- Penalize saturated edges
- Next lab

```
Maze
```

Maze escape

Browser lab track



Browser lab starter

Implement track

- Implement `z_route(a, b, prefer)`
- Print Z and L paths for the same terminal pair; explain one edge difference

Pitfalls

- Using a midpoint that leaves zero-length segments
- Not falling back to L on aligned pins
- Confusing Z prefer HZ with L prefer HV naming

Your turn

- Finish Z routing on two-pin nets
- Next: maze routing that respects congested edges